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Port P3 pins P3.6 and P3.7 of IC1 are configured to select the control source for the devices. These are connected to DIP switches S17 and S18 and pulled high via resistors R2 and

R1, respectively. Here, we are using two control sources, switches and mobile's key. DIP switches S17 and S18 select the control sources as shown in Table II.

TABLE II
Control Source Selection Using DIP Switches

S17	S18	Control status
0	0	Rest status
0	1	Switches only
1	0	Mobile only
1	1	Switches and mobile

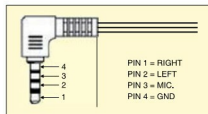


Fig. 2: Pin configuration of mobile headset

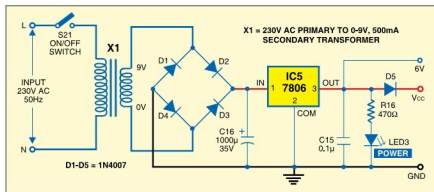


Fig. 3: Power supply circuit

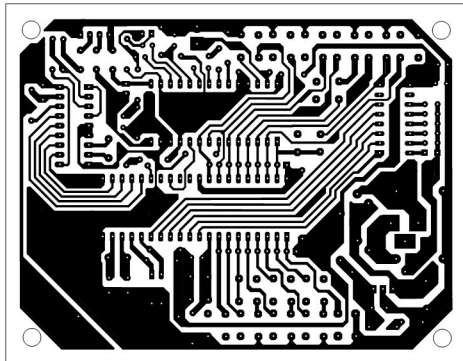


Fig. 4: A single-side, actual-size PCB layout for cellphone-based device control with voice acknowledgement

Pin 2.5 of Port P2 is configured to show the rest status. That is, if none of the control sources is selected by DIP switches S17 and S18, LED1 glows. Resistor R14 limits the current through LED1.

Voice acknowledgement is provided by the APR9600 (IC2). It is a single-chip voice recording and playback device that can record and play multiple messages at random or in sequential mode for 60 seconds. The user can select sample rates with corresponding-quality recording lengths. Microphone amplifier, automatic gain control (AGC) circuits, internal anti-aliasing filter, internal output amplifier and message management are some of the features of the APR9600.

Here the APR9600 is configured in random-access mode, which supports two, four and eight messages of fixed durations. The length of each message is the total recording length available divided by the total number of memory segments/tracks enabled.

Audio processor APR9600 can store up to eight voice messages. Port P0 pins and P2.7 are configured to communicate with IC2. Port P0 pins trigger selection of the message. Port pin P2.7 is the input signal to identify whether the voice message is playing or not.

Pins P3.0 through P3.5 of Port P3 control the devices with the help of relays RL1 through RL6 via relay driver IC4.

A speaker is connected to IC2 for audio output. The speaker output drives the mic input of the mobile for audio acknowledgement. An electret microphone MIC1 is connected to IC2 to record the voice in IC2. LED2 flashes to show the busy status of IC2 during recording and playback. The audio messages to be recorded in APR9600, by using trigger switches S9 through S16, are shown in Table III. SPST switch S19 is closed for recording and switch S19 is opened for playback.

Fig. 3 shows the power sup-